

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	18 June 2026
Student Name	Narendula Jashwanth
Project Name	ShopEZ – Full Stack E-Commerce Platform Using MERN Stack
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the frontend and backend project structure.	2	High	Jashwanth
Sprint-1	User Registration	USN-2	As a user, I can register using name, email, and password.	3	High	Jashwanth
Sprint-1	User Login	USN-3	As a user, I can log in using email and password.	3	High	Jashwanth
Sprint-1	Authentication	USN-4	As a user, my login details are secured using authentication.	3	High	Jashwanth
Sprint-1	User Profile	USN-5	As a user, I can view and manage my profile details	2	Medium	Jashwanth
Sprint-1	Database	USN-6	As a developer, I can create the users collection in the database.	2	High	Jashwanth
Sprint-1	UI Layout	USN-7	As a user, I can view a simple homepage and navigation bar.	3	High	Jashwanth
Sprint-1	Testing	USN-8	As a developer, I can test registration and login APIs.	2	High	Jashwanth
Sprint-2	Product Management	USN-9	As a user, I can view the product list.	3	High	Jashwanth
Sprint-2	Product Details	USN-10	As a user, I can view complete product details.	2	High	Jashwanth

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Admin Product	USN-11	As an admin, I can add new products.	3	High	Jashwanth
Sprint-2	Admin Product	USN-12	As an admin, I can update or delete products.	3	High	Jashwanth
Sprint-2	Cart System	USN-13	As a user, I can add products to cart.	3	High	Jashwanth
Sprint-2	Cart System	USN-14	As a user, I can update quantity or remove products from cart.	3	High	Jashwanth
Sprint-2	Cart Total	USN-15	As a user, I can view the total cart amount.	2	High	Jashwanth
Sprint-2	Database	USN-16	As a developer, I can create products and cart collections.	1	High	Jashwanth
Sprint-3	Order Management	USN-17	As a user, I can place an order from the cart.	4	High	Jashwanth
Sprint-3	Order History	USN-18	As a user, I can view my order history.	3	High	Jashwanth
Sprint-3	Order Status	USN-19	As a user, I can check order status.	2	Medium	Jashwanth
Sprint-3	Admin Dashboard	USN-20	As an admin, I can view the dashboard.	3	High	Jashwanth
Sprint-3	Admin Orders	USN-21	As an admin, I can manage customer orders.	3	High	Jashwanth
Sprint-3	Admin Users	USN-22	As an admin, I can view user details.	2	Medium	Jashwanth
Sprint-3	Database	USN-23	As a developer, I can create the orders collection.	2	High	Jashwanth
Sprint-3	Testing	USN-24	As a developer, I can test order-related APIs.	1	High	Jashwanth
Sprint-4	Integration	USN-25	As a developer, I can connect frontend, backend, and database.	4	High	Jashwanth
Sprint-4	Validation	USN-26	As a user, I can see proper error and success messages.	3	High	Jashwanth
Sprint-4	UI Improvement	USN-27	As a user, I can use a clean and responsive interface.	3	Medium	Jashwanth
Sprint-4	Final Testing	USN-28	As a developer, I can test all modules completely.	4	High	Jashwanth

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	Deployment	USN-29	As a developer, I can prepare the project for deployment.	2	Medium	Jashwanth
Sprint-4	Documentation	USN-30	As a developer, I can prepare final project documentation.	2	High	Jashwanth
Sprint-4	Screenshots	USN-31	As a developer, I can add screenshots and final output images.	2	Medium	Jashwanth

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	15 Feb 2026	20 Feb 2026	20	20 Feb 2026
Sprint-2	20	6 Days	21 Feb 2026	26 Feb 2026	20	26 Feb 2026
Sprint-3	20	6 Days	27 Feb 2026	04 Mar 2026	20	04 Mar 2026
Sprint-4	20	6 Days	05 Mar 2026	10 Mar 2026	20	010Mar 2026

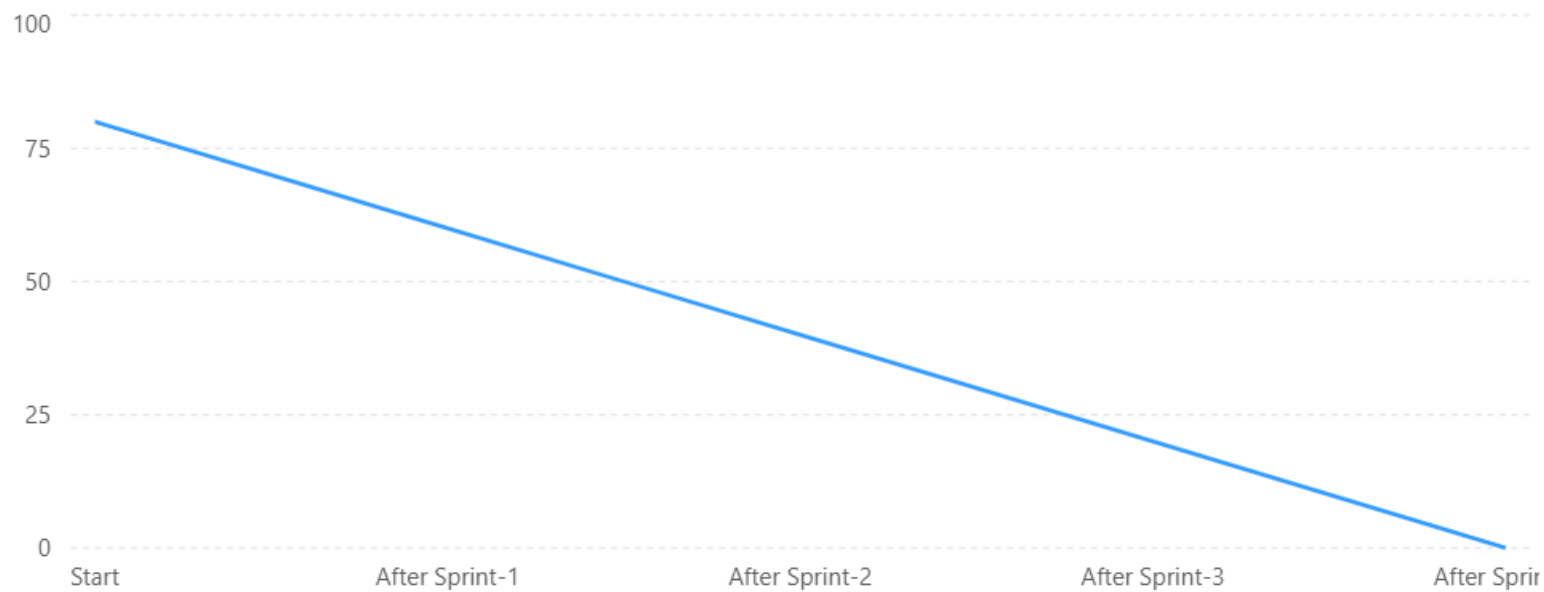
Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Total Story Points = 80
Number of Sprints = 4
Average Velocity = Total Story Points / Number of Sprints
Average Velocity = $80 / 4$
Average Velocity = 20 story points per sprint
Sprint Duration = 6 Days
Average Velocity per Day = $20 / 6$
Average Velocity per Day = 3.33 story points per day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



The burndown chart shows steady progress in the ShopEZ individual project. The planned work is completed within four sprints, covering authentication, product management, cart system, order management, admin dashboard, testing, and documentation.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
<https://www.atlassian.com/agile/tutorials/epics>
<https://www.atlassian.com/agile/tutorials/sprints>
<https://www.atlassian.com/agile/project-management/estimation>
<https://www.atlassian.com/agile/tutorials/burndown-charts>